

- 7 As the human population in the world has increased, the demand for food has increased.

The production of food has an impact on the environment, unless it can be done in a sustainable way.

- (a) (i) State the meaning of the term **population**, as it has been used in this context.

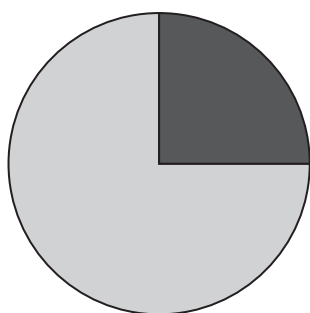
(1)

- (ii) State the meaning of the term **sustainable**, as it has been used in this context.

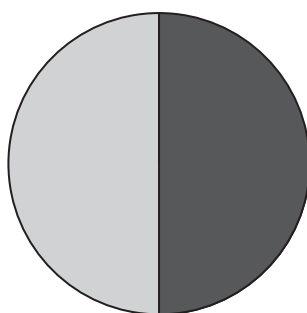
(1)

- (b) The diagram shows the impact of food production on three environmental factors.

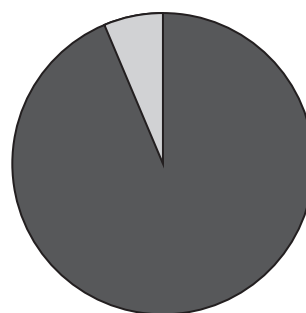
Greenhouse gases



Habitable land use



Biodiversity of mammals



not related to food production



related to food production

(i) Explain how food production could contribute to the greenhouse effect.

(3)

(ii) Land occupies 149 million km^2 of the surface of the Earth.

Habitable land is 71% of this area and some of it is used for food production.

Calculate the area of the surface of the Earth used in food production.

Express your answer in standard form.

(3)

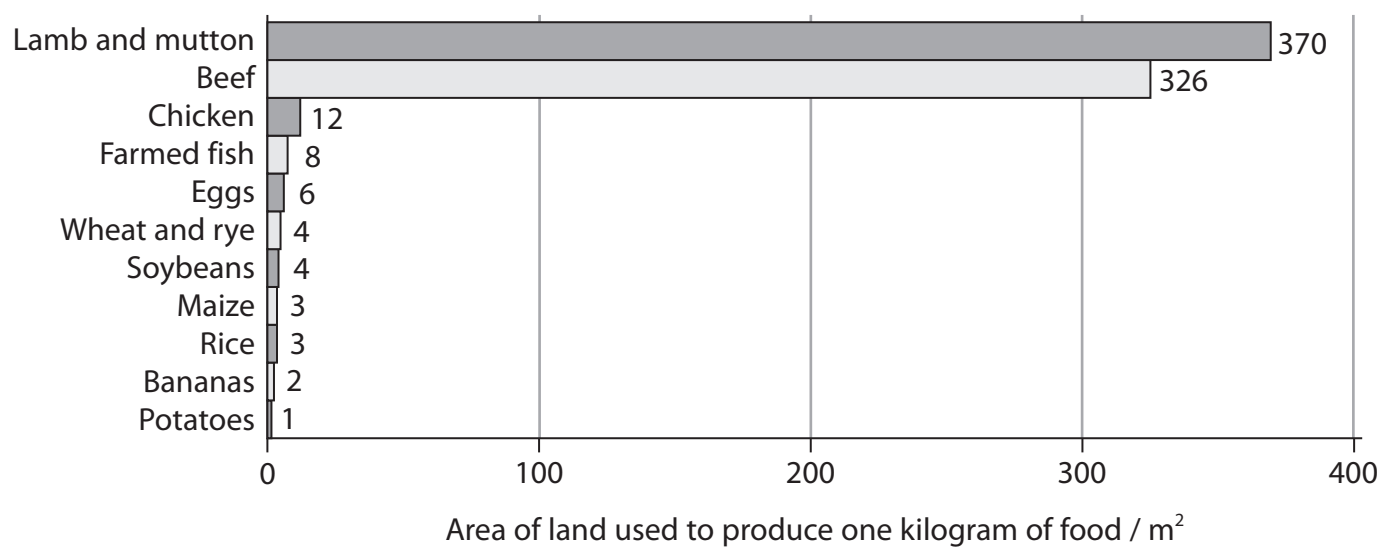
Answer km^2

(iii) Estimate the effect of food production on the ratio of biodiversity of mammals.

(1)

Answer

*(c) The diagram shows the area of land used to produce one kilogram of some food products.



Explain how food production could be made more sustainable.

Use the information in the diagram and your own knowledge to support your answer.

(6)

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(Total for Question 7 = 15 marks)